

Simulation of swimming oblate jellyfish with a paddling-based locomotion

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The hydrodynamics of a swimming jellyfish depends on the morphology of the species. For example, oblate jellyfish appear to generate wide vortex structures near the bell margin. The vortex structures affect both the propulsion system and the feeding structure because the swimming and prey capturing activities are interrelated processes in these taxa. A three-dimensional computational model was established for an oblate jellyfish to analyse how the vortex structures present in the wake affect the swimming mechanism and the propulsion efficiency, which is defined as the ratio of power output (thrust multiplied by centre velocity) to power input (energy rate required for bell contraction). An improved penalty immersed boundary method was adopted to consider the interactions between the swimming jellyfish and the ambient fluid. The vortex structures in the wake of the swimming jellyfish were investigated in detail. The vortices generated by the contraction and expansion of the jellyfish bell interact with the vortex structures generated by the forward-moving behaviour of the jellyfish. The resulting vortex structures not only transfer momentum to the swimming jellyfish via the fluid, thereby providing the main source of thrust, but also have an implication for feeding. The effects of the elastic properties of the jellyfish on the propulsion were examined. The propulsion efficiency reaches its optimum value at particular elastic properties. We also investigated the effect of the swimming pattern of jellyfish on the propulsion efficiency. The efficiency increases with the flapping frequency and force duration.

Key words: propulsion, swimming/flying, vortex dynamics

1. Introduction

Interactions between flying or swimming animals and the surrounding fluid are essential to flight or swimming movements, and yet the dynamics governing these interactions is extremely complex and difficult to elucidate. Research into animal–fluid interactions and biological propulsion mechanisms has been conducted extensively since the dynamics associated with animal–fluid interactions is important for the energetics of the movement. Animals have tuned their movement properties to

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facilitate feeding (Lima 1985; Summers *et al.* 1998; Dabiri *et al.* 2005), to minimize the time exposed to predators (Lima 1985) or to optimize propulsive efficiency (Liao *et al.* 2003; Taylor, Nudds & Thomas 2003; Ulrike 2003; Dabiri *et al.* 2005). Animals often want to swim for efficiency, maximum speed, feeding, escape manoeuvres or stealth depending on various circumstances. Optimal propulsion conditions may be reached by modulating the flapping frequency and the properties of the vortex structures generated by a flapping motion. Fish swimming behind a stationary object can exploit the vortices that form in the wake of the object, thereby modifying their swimming behaviour to minimize locomotion costs (Liao *et al.* 2003; Ulrike 2003). Apart from optimization of their swimming efficiency, fish are able to accelerate and turn by generating individual vortex rings (Linden & Turner 2004). Salps produce optimized vortex rings directly by ejecting the fluid through a rear orifice (Linden & Turner 2004). Dabiri *et al.* (2005) examined the relevance of the vortex generated in the wake of a swimming jellyfish to propulsion. Taylor *et al.* (2003) recently showed that the flapping frequency, flapping amplitude and forward velocity are related to the propulsive efficiency in a variety of swimming or flying animals.

Swimming jellyfish were the earliest animals to have generated thrust based on their own muscle power (Costello, Colin & Dabiri 2008). The swimming cycle of a jellyfish consists of a contraction phase (i.e. a power stroke), a relaxation phase (i.e. a recovery stroke) and a coasting phase (i.e. motion under inertia). Animal–fluid interactions such as those present in jellyfish-in-water systems pose several challenges to experimental studies due to difficulties associated with measuring the moving boundaries of jellyfish. Despite these challenges, a variety of experimental studies have revealed the rich phenomena that underlie the swimming efficiency and mechanism. Daniel (1985) experimentally examined the propulsive efficiency of jellyfish by relating the energy consumed by the muscle to the forward velocity. Some planktonic medusae generate thrust primarily by a jet propulsion mechanism derived from the subumbrella volume change that occurs during the contraction and relaxation phases and, thus, are termed ‘jetters’ (Colin & Costello 2002). In these taxa it is suspected that the swimming mechanism is decoupled from the feeding structure (Colin, Costello & Klos 2003). Recently, a second swimming mechanism was identified in which jellyfish achieve a locomotive force using a paddling motion on the bell margin and these taxa are termed ‘rowers’ (Colin & Costello 2002; Mchenry & Jed 2003; Costello *et al.* 2008). These medusae including hydromedusae and scyphomedusae generally have oblate bell morphologies and capture prey simultaneously while swimming (Colin *et al.* 2003). The hydrodynamic patterns underlying the paddling swimming mechanism are not as simple as the patterns found in the jet swimming mechanism. The contraction phase of jet motion is characterized by the formation of first a single vortex ring, and then a strong jet in the wake of the swimming jellyfish (Costello *et al.* 2008). The vortex ring is called the ‘starting vortex’ (Dabiri *et al.* 2005). In paddling motion, on the other hand, the starting vortex generated during the contraction phase is followed by a second vortex ring called the ‘stopping vortex’ (Dabiri *et al.* 2005), which is generated during the relaxation phase. The starting vortex interacts with a subsequently formed stopping vortex and has a rotational direction opposite to that of the stopping vortex. The stopping vortex plays an important role in swimming mechanisms (Costello *et al.* 2008).

Many experimental studies have attempted to identify the relationship between the swimming mechanism and the vortex structures in the wake (Dabiri *et al.* 2005; Costello *et al.* 2008; Dabiri *et al.* 2009). The vortex structures depend not only on the swimming mechanism but also on the patterns determined by the pulse intervals

(Megill 2002). Within the duration of each phase in a swimming mechanism, different swimming patterns are produced that may be useful in a variety of circumstances, such as fishing, travelling or escaping (Mills 1981; Megill 2002). Santhanakrishnan *et al.* (2012) investigated the hydrodynamics in the wake of the *Cassiopea* jellyfish for different pulse intervals. The elastic properties of a jellyfish affect the propulsion and the swimming mechanism because the force required to refill the bell depends on the elastic energy stored in the bell during the contraction phase (Demont & Gosline 1988*a,b,c*; Megill, Gosline & Blake 2005). Analytical models have been developed and compared with experimental data by combining the morphometric and kinematic data from jellyfish (Daniel 1983, 1985; Colin & Costello 2002; Dabiri & Gharib 2003; Mchenry & Jed 2003). Analytical models predict the kinematic data, such as the forward velocity and acceleration, based on the morphometric information about a swimming jellyfish, such as the bell volume change, height and diameter. The added mass due to the inertial effects of the surrounding fluid during acceleration and deceleration of the immersed body is also described well by analytical models (Daniel 1985; Colin & Costello 2002; Dabiri & Gharib 2003). Analytical models have been found to be limited, however, in their ability to predict the hydrodynamic patterns, such as the vortex structures generated.

Dular, Bajcar & Sirok (2009) modelled the hydrodynamics of swimming jellyfish using the SIMPLE algorithm (Patankar 1980). The elastic properties of the jellyfish bell were not considered in this model, and the jellyfish motion was instead described using boundary conditions. Sahin & Mohseni (2009) adopted the arbitrary Lagrangian–Eulerian (ALE) formulation to study the flow patterns generated by an oblate jellyfish. The elasticity was assumed to be linear in the Cartesian coordinate system, in an effort to mimic the elasticity of the jellyfish bell. In this simulation, the vortex structures in the wake were investigated in a two-dimensional computational domain, and they took a form similar to that described by Dabiri *et al.* (2005). In addition to the ALE method, researchers have used the immersed boundary method (IBM) to model the propulsion of swimming jellyfish in two-dimensional simulations (Zhao, Freund & Moser 2008; Huang & Sung 2009; Herschlag & Miller 2011). Zhao *et al.* (2008) and Huang & Sung (2009) used the body force to mimic the contraction and expansion kinematics, while the jellyfish bell was deformed through the prescribed motions in Herschlag & Miller (2011). This study offered a detailed investigation of the influences of the Reynolds number and jellyfish morphology on the propulsion mechanism and hydrodynamics. Alben, Miller & Peng (2013) simulated jet-propelled jellyfish (jetters) with an axisymmetric bell shape to investigate the relationship between kinematics and performance, in which the bell deformed with the prescribed motion. Rudolf & Mould (2010) simulated swimming jellyfish in two dimensions and then extrapolated the results to three dimensions; however, this study only concentrated on the motions of a swimming jellyfish. Thus far, few numerical studies have examined the propulsive efficiency and hydrodynamics of various swimming gaits in three dimensions.

The present study simulated the hydrodynamics of a three-dimensional swimming jellyfish in a quiescent fluid. In three-dimensional simulations, a realistic bell model can be obtained since the bell deformation is constrained in the circumferential direction by circumferential elastic properties, unlike in two-dimensional or axisymmetric simulations. The body force and the mechanism underlying the elastic energy storage required to contract and refill the jellyfish bell were considered simultaneously in the simulation. The jellyfish motion equations, which take into account the elasticity of the bell, were solved using the subdivision finite element method (FEM) on a

moving Lagrangian grid (Cirak, Ortiz & Schroder 2000; Huang, Chang & Sung 2012). On the other hand, the fluid motion equations were solved using the fractional step method on a fixed Eulerian grid (Kim, Baek & Sung 2002). The fluid–structure interactions were simulated by adopting the penalty immersed boundary method (pIBM) developed by Huang, Chang & Sung (2011), in which the solid and fluid motion equations are coupled by adding a momentum force term. The main purpose of the present study was to investigate the hydrodynamic patterns generated near the bell of an oblate jellyfish using a paddling-based locomotion. The interactions between the vortex structures generated by both the strokes of the jellyfish and the forward motions were considered. This approach is distinct from previous studies in which only the effects of the strokes were examined (Costello *et al.* 2008; Dabiri *et al.* 2005, 2009). The effects of the jellyfish elasticity on the swimming performance were studied by varying the elastic parameters. The effects of the swimming patterns on the vortex structures in the wake were also investigated. We focused on predicting the propulsion efficiency in terms of the Froude efficiency, as used widely (Mchenry, Pell & Long 1995; Liu, Wassersug & Kawachi 1996; Sahin, Mohseni & Colin 2009). The Froude propulsive efficiency, which is determined by the ratio of the power output obtained by the thrust multiplied by the centre velocity to the power input obtained by measuring the rate of energy required for the bell deformation, was examined with a view to investigating the elastic characteristics and swimming patterns. In § 2, we introduce the formulation for the simulation of a swimming jellyfish in a quiescent fluid, and we briefly describe the numerical method. The numerical results and discussion are presented in § 3. Finally, a summary is given in § 4.

2. Problem formulation

We describe a jellyfish bell as a hemiellipsoid with a measurable aspect ratio. This model was first used for hydrodynamic analysis studies by Daniel (1983, 1985). In the present study, we considered a 3D oblate medusa with a bell diameter and height of 1.0 and 0.5, respectively, with a fineness ratio of 0.5, as defined by the ratio of the bell height to the diameter. The bell was defined as a membrane without thickness and subject to bending effects. Figure 1(a) shows a schematic of the computational domain and the hemispheroidal bell used in the simulation. The animal–fluid interactions were modelled by solving the solid and fluid motion equations separately, and these equations were coupled using the pIBM.

The Lagrangian variables (s_1, s_2) in the solid motion equation were defined on a moving curvilinear grid system. The position, denoted by $\mathbf{X}(s_1, s_2, t)$, depends on

$$\rho \frac{\partial^2 \mathbf{X}}{\partial t^2} = \mathbf{F}_e - \mathbf{F}_l + \mathbf{F}_{ex}, \quad (2.1)$$

where ρ denotes the difference between the densities of the solid and the fluid, $\mathbf{F}_e$ is the elastic force term describing the elastic characteristics of the jellyfish membrane, $\mathbf{F}_l$ is the Lagrangian momentum force that characterizes the fluid–membrane interactions and $\mathbf{F}_{ex}$ denotes the body force required during the contraction and relaxation phases of swimming. The elastic force term $\mathbf{F}_e$, obtained using the energy method (Fung & Tong 2001), takes the form

$$\mathbf{F}_e = \sum_{i,j=1}^2 \left[\frac{\partial}{\partial s_i} \left(\mathbf{s}_{ij} \frac{\partial \mathbf{X}}{\partial s_j} \right) - \frac{\partial^2}{\partial s_i \partial s_j} \left(\mathbf{B}_{ij} \frac{\partial^2 \mathbf{X}}{\partial s_i \partial s_j} \right) \right], \quad (2.2)$$

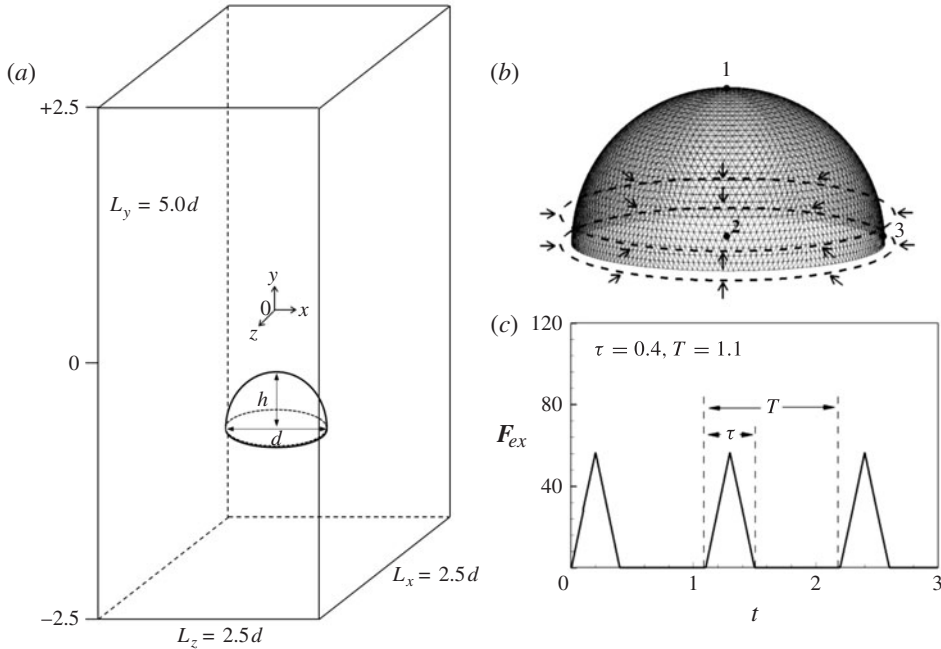


FIGURE 1. (a) Schematic diagram of the jellyfish bell in a quiescent fluid and the computational domain. (b) Schematic diagram showing the triangular mesh and the body force applied to the jellyfish bell. Points 1, 2 and 3 denote the top point, the representative centre point (the height of which is the same as the heights of the force points) and the force point, respectively. (c) Time histories of the body force required to contract the bell of the triangular mesh and the body force applied on the jellyfish bell.

which contains the second Piola–Kirchhoff stress tensor $\mathbf{S}_{ij}$ and the bending stress tensor $\mathbf{B}_{ij}$.

The stress tensors $\mathbf{S}_{ij}$ and $\mathbf{B}_{ij}$ could be obtained by deriving the elastic force through the principle of virtual work (Huang *et al.* 2012). The shearing stress tensors were described using Skalak's law because the shearing elastic characteristics of the jellyfish bell were similar to the characteristics of a membrane defined by Skalak's law in which the stress–strain curve displays strain hardening under large deformations (Barthes-Biesel, Diaz & Dhenin 2002). The shearing stress tensors of a Skalak membrane are composed of the Green strain tensor and its invariants, as follows:

$$\mathbf{S}_{ij} = \frac{\varphi}{2} \left[(I_1 + 1)\delta_{ij} + (-1)^{i+j}(CI_2 - 1)(\delta_{ij} + 2D_{3-i,3-j}) \right] \quad (i, j = 1, 2), \quad (2.3)$$

where φ denotes the tension modulus, φC is the area dilation modulus, and I_1 and I_2 are the alternate forms of the first and second invariants of the Green strain tensor,

$$\left. \begin{aligned} I_1 &= 2(D_{11} + D_{22}), \\ I_2 &= 2(D_{11} + D_{22}) + 4(D_{11}D_{22} - D_{12}D_{21}). \end{aligned} \right\} \quad (2.4)$$

The Green strain tensor takes the form

$$\mathbf{D}_{ij} = \frac{1}{2} \left(\frac{\partial \mathbf{X}}{\partial s_i} \cdot \frac{\partial \mathbf{X}}{\partial s_j} - \frac{\partial \mathbf{X}^0}{\partial s_i} \cdot \frac{\partial \mathbf{X}^0}{\partial s_j} \right) \quad (i, j = 1, 2), \quad (2.5)$$

where $\mathbf{X}^0$ denotes the initial position of the membrane. The bending stress tensors can be expressed in terms of the bending strain tensor $\mathbf{K}_{ij}$ as

$$\mathbf{B}_{ij} = \zeta [(K_{11} + K_{22})\delta_{ij} - (-1)^{i+j}(1 - \nu_s)K_{3-i,3-j}] / \left(\frac{\partial^2 \mathbf{X}}{\partial s_i \partial s_j} \cdot \frac{\partial^2 \mathbf{X}}{\partial s_i \partial s_j} \right)^{1/2} \quad (i, j = 1, 2), \quad (2.6)$$

where ζ denotes the bending coefficient and ν_s denotes the Poisson ratio. The bending strain tensor $\mathbf{K}_{ij}$ is expressed by

$$\mathbf{K}_{ij} = \left(\frac{\partial^2 \mathbf{X}}{\partial s_i \partial s_j} \cdot \frac{\partial^2 \mathbf{X}}{\partial s_i \partial s_j} \right)^{1/2} - \left(\frac{\partial^2 \mathbf{X}^0}{\partial s_i \partial s_j} \cdot \frac{\partial^2 \mathbf{X}^0}{\partial s_i \partial s_j} \right)^{1/2} \quad (i, j = 1, 2). \quad (2.7)$$

In the limit of small deformation the Skalak law is reduced to Hooke's law, where the Poisson ratio is $C/(1 + C)$ (Barthes-Biesel *et al.* 2002). In the present study, C was set to 100 for area incompressibility of the jellyfish bell and the Poisson ratio was almost equal to unity. However, the Poisson ratio involved in $\mathbf{B}_{ij}$ was set to zero, which is considered to be independent of that implied by the shearing stress tensor $\mathbf{S}_{ij}$. This is reasonable in cell membrane, where different mechanisms are applied to the in-plane tension and the bending moment (Huang *et al.* 2012).

The body force, denoted by $\mathbf{F}_{ex}$, represents the action of the swimming muscle located along the bell. In the case of the oblate jellyfish, bell variations generally occur only near the bell margin (Ford & Costello 2000). Accordingly, we applied the body force under the condition that the distance from the forcing point to the bell top was larger than 0.5 (the range with body force occupies approximately 33 % of the total arclength of the bell). The body force was applied to the membrane in the form of a nodal force in the radial direction, as depicted in figure 1(b). The body force was described as

$$\mathbf{F}_{ex} = \begin{cases} \alpha |\mathbf{X}_{13}| \min(t, \tau - t) \frac{\mathbf{X}_{23}}{|\mathbf{X}_{23}|}, & |\mathbf{X}_{13}| > 0.5, t < \tau, \\ 0, & \text{otherwise,} \end{cases} \quad (2.8)$$

where α denotes the body force constant, $|\mathbf{X}_{13}|$ is the distance between the bell top point and each force point, τ is the force duration and $\mathbf{X}_{23}$ is the vector directed from the force point to the centre point of the cross section drawn at the same height from the top point (Huang & Sung 2009). The body force constant α was set to be 400 when the force duration was 0.4. This constant varied with the force duration to ensure that the same impulse of the body force was applied to the jellyfish in the following simulations. Figure 1(c) shows the time history of the body force depicted by (2.8). The flapping frequency f_{fl} is defined as the inverse of the flapping period T .

The fluid motion is described in a fixed Cartesian coordinate system $Oxyz$ in which the origin is located at the domain centre (figure 1a). The fluid motion is governed by the Navier–Stokes equations and the continuity equation, as expressed by

$$\rho_0 \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{f}, \quad (2.9)$$

$$\nabla \cdot \mathbf{u} = 0, \quad (2.10)$$

where ρ_0 is the fluid density, $\mathbf{u}$ is the velocity vector, p is the pressure, μ is the dynamic viscosity and $\mathbf{f}$ is the Eulerian momentum. The non-dimensional forms

of the governing equations were derived for use in the computations. The fluid density ρ_0 was adopted as the characteristic density, the bell diameter d as the characteristic length and the maximum velocity v as the characteristic velocity. The characteristic parameters determined the following characteristic scales: d/v for time, $\rho_0 v^2$ for pressure p , $\rho_0 v/d$ for the Eulerian momentum $\mathbf{f}$, $\rho_0 v^2 d$ for the second Piola–Kirchhoff stress tensor $\mathbf{S}_{ij}$ and the tension modulus φ , $\rho_0 v^2 d^3$ for the bending stress tensor $\mathbf{B}_{ij}$ and the bending coefficient ζ , and $\rho_0 v^2$ for the Lagrangian momentum $\mathbf{F}_l$. The Navier–Stokes equations, derived in terms of characteristic scales, took the non-dimensional form

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = -\nabla p + \frac{1}{Re} \nabla^2 \mathbf{u} + \mathbf{f}, \quad (2.11)$$

where $Re = \rho_0 v d / \mu$. For convenience, the same notations were used in (2.11) for the dimensionless variables as their dimensional counterparts. The non-dimensional forms of the continuity and solid motion equations were the same as the dimensional ones. It should be noted that the jellyfish bell was assumed to have no thickness, thus ρ in (2.1) is a surface density and its non-dimensional form is expressed as $\rho = (\rho_1 - \rho_0) c / \rho_0 d$, with ρ_1 the volume density of the jellyfish and c its thickness.

The pIBM was implemented to deal with the interactions between the fluid flow and the solid motion. In the pIBM, the immersed boundary was separated into two parts: a massless part that interacted with the local fluid directly and a massive part related to the membrane motion. The two parts were linked by a stiff spring with damping. The restoring force of the spring and damping system was generated between the fluid and the membrane and was added to (2.1) in the form of $-\mathbf{F}_l$, as expressed by

$$\mathbf{F}_l(s_1, s_2, t) = -\kappa[(X_{ib} - \mathbf{X}) + \Delta t(\mathbf{U}_{ib} - \mathbf{U})], \quad (2.12)$$

where κ is a large constant, Δt denotes the computational time step, X_{ib} and $\mathbf{U}_{ib}$ are, respectively, the position and velocity of the massless part, and $\mathbf{X}$ and $\mathbf{U}$ are the position and velocity of the massive part. The position of the massive part $\mathbf{X}$ was determined by the membrane motion equation (2.1), and the following velocity was derived based on the relation between the position and velocity, i.e. $\mathbf{U} = \partial \mathbf{X} / \partial t$. The velocity $\mathbf{U}_{ib}$ of the massless part was derived by interpolating the local Eulerian fluid velocity using the smoothed Dirac delta function $\delta()$, as follows:

$$\mathbf{U}_{ib} = \int_{\Omega_f} \mathbf{u}(\mathbf{x}, t) \delta(\mathbf{X}(s_1, s_2, t) - \mathbf{x}) d\mathbf{x}. \quad (2.13)$$

The velocity of the massless part was used to determine the position according to the equation

$$\mathbf{X}_{ib} = \mathbf{X}_{ib}^0 + \int_0^t \mathbf{U}_{ib} dt, \quad (2.14)$$

where $\mathbf{X}_{ib}^0$ denotes the initial position of the massless part (as well as of the massive part). The counterpart of the restoring force $\mathbf{F}_l$, which is defined on the Lagrangian domain, is denoted by $\mathbf{f}$ and is obtained by spreading the restoring force to the Eulerian domain using the smoothed Dirac delta function as follows:

$$\mathbf{f}(\mathbf{x}, t) = \int_{\Omega_s} \mathbf{F}_l(s_1, s_2, t) \delta(\mathbf{x} - \mathbf{X}(s_1, s_2, t)) ds_1 ds_2. \quad (2.15)$$

In the frame of the pIBM, the fluid solver and the solid solver were developed separately by adding the momenta $\mathbf{f}$ and $\mathbf{F}_l$ to each governing equation. The fluid

motion equation was solved by the fractional step method (Kim *et al.* 2002) and the solid motion equation was solved by the subdivision FEM (Cirak *et al.* 2000; Huang *et al.* 2012). Details of the numerical procedure were described in Huang *et al.* (2012).

3. Results and discussion

We examined the behaviour of the swimming jellyfish in a quiescent fluid, as schematically shown in figure 1(a). The rectangular box indicates the fluid computational domain, with a size of $2.5 \times 5.0 \times 2.5$ in the transverse (x), streamwise (y) and spanwise (z) directions, respectively. The domain was scaled by the bell diameter. Neumann boundary conditions were applied at the boundaries. A uniform grid, $128 \times 256 \times 128$ in size, was oriented along the x , y and z axes. The jellyfish bell was discretized into 6144 triangular elements, and the element size was comparable with the mesh size in the fluid domain. The mesh was refined on an oblate jellyfish bell by applying a quadrissection refinement procedure to an initially coarse triangular mesh. The nodal positions of the refined mesh were determined based on the points of the old mesh by using the subdivision surface method (Cirak *et al.* 2000). A smooth mesh was obtained on the bell by repeating the above process five times, as shown in figure 1(b). In the following subsections, the results are presented in three sections. Section 3.1 offers an explanation of the hydrodynamics surrounding swimming jellyfish in view of the vortex structures. Section 3.2 describes how the elastic properties of the bell affect the swimming mechanism and propulsive efficiency. Finally, we discuss the effect of the swimming pattern on the swimming efficiency.

3.1. Swimming mechanism

Prior to conducting the simulations over a range of parameters, we validated the numerical method by comparing the simulation results with experimental results reported previously. Figure 2 shows a comparison of the time histories of the centre velocity determined using the present simulation method and the experimental data collected by Mchenry & Jed (2003). The non-dimensional forms of the equations were solved, as indicated in § 2, and the results were transformed into the dimensional data shown in figure 2 for comparison with the experimental data. The Reynolds number was set to 500, based on the jellyfish (*Aurelia aurita*) examined in the experimental study. The tension modulus φ was set to 180, which was within the range of reasonable values measured in the experiments of Demont & Gosline (1988a) and Megill *et al.* (2005). A bending modulus of $\xi = 0.6$ was selected for (2.6) based on a comparison with the experimental data. The non-dimensional density difference was set to $\rho = 0.5$, which is a large value primarily due to numerical stability considerations. Although it is generally higher than the real case, it is considered to be qualitatively valid as estimated from the phenomenon of sink fishing, a behaviour in which the jellyfish extends its tentacles completely and sinks due to the negative buoyancy even though the tentacles and oral arms play significant roles in increasing the drag. Huang & Sung (2009) examined the effect of the density difference on the forward motion of a 2-D swimming jellyfish. They showed that the time histories of the centre velocity differ slightly and that for a large density difference lags more due to the inertial force. The result converged to that of the neutrally buoyant case as the density difference approached zero. A computational time step of $\Delta t = 0.000\,03$ and a free constant of $\kappa = 2 \times 10^7$ were selected for calculation of the Lagrangian force based on (2.12).

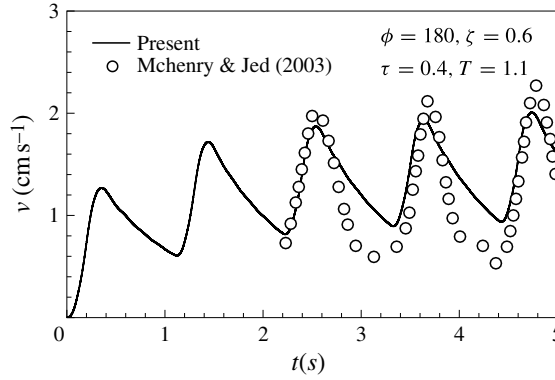


FIGURE 2. Time hisotries of the centre velocity and a comparison of our simulation results with measured experimental data for *A. aurita*.

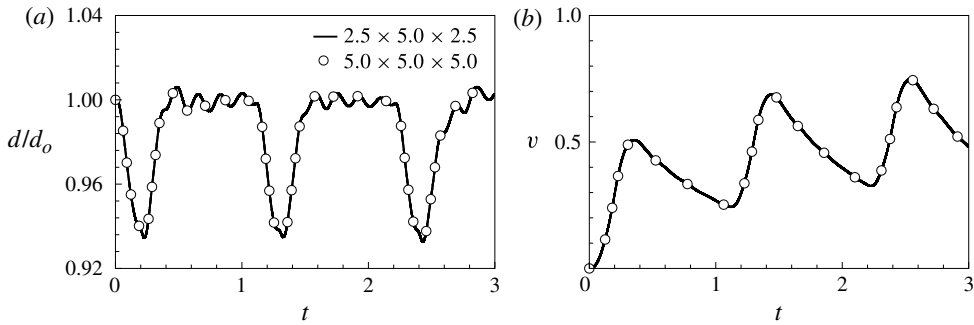


FIGURE 3. Time histories of (a) the bell diameter variations and (b) the centre velocity for different domain sizes.

When the force duration τ and the period T were set to values similar to those corresponding to the experimental conditions, similar trends in the forward centre velocity were observed, as shown in figure 2. The parameters listed above were also used to examine whether the computational domain was sufficiently large in the transverse (x) and spanwise (z) directions. Figure 3 shows that the time histories of the bell diameter and the centre velocity agreed well over two domain sizes, indicating that the computational domain used in the simulation was sufficiently large.

The model was validated based on the representative kinematics shown in figure 4. The centre position in the forward-moving direction increased monotonically, as shown in figure 4(a). The time history of the fineness ratio is shown in figure 4(b). This ratio oscillated after the power and recovery strokes due to the elastic energy stored in the contraction phase. Figure 4(c) shows the time history of the bell diameter normalized by its initial value. Characteristic instants (A–F) during the bell stroke are indicated on the curve. The power stroke began at point A and ended at point C. The recovery stroke then began at point C and persisted until point F. During the contraction phase, points A–C, the centre velocity increased, as shown in figure 4(d). As the recovery stroke began at point C, the centre velocity still increased, indicating that jet propulsion alone was not sufficient to explain the swimming mechanisms of the oblate jellyfish since the thrust occurred only during the contraction period in jet

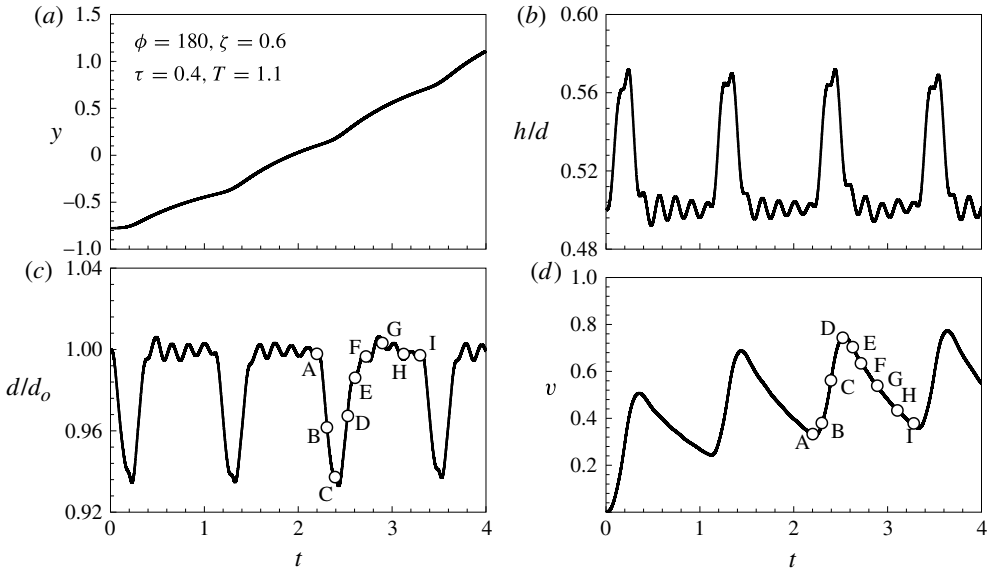


FIGURE 4. Time histories of representative kinematic measures of the swimming jellyfish: (a) centre position; (b) fineness ratio; (c) rate of change of the bell diameter; (d) centre velocity. The points indicating the bell deformations and the centre velocity are marked. Point A indicates the beginning of the contraction, point B indicates the middle of the contraction, point C indicates the end of the contraction, point D indicates the point at which the centre velocity reaches a maximum value during the propulsive cycle. Points E and F indicate the end of the recovery stroke.

motion. The centre velocity reached a maximum at point D, after which point the velocity decreased until the start of the next power stroke, as the drag force applied by the surrounding fluid exceeded the thrust.

Johnson & Patel (1998) investigated the flow of an incompressible viscous fluid past a fixed sphere numerically and experimentally, with the result of axisymmetric flow in the wake for a Reynolds number of up to 200. In the present simulation, the hemisphere describing swimming jellyfish that have oblate bell morphology (fineness ratio of 0.5) generates vortex structures not only by the strokes but also by the forward-moving behaviour. The flow in the wake shed from the shear layer by the forward motion of the bell takes a similar form to the axisymmetric flow behind the sphere. The vortex structures and the velocity vectors in the central symmetric plane at each of the characteristic instants (A–I) are displayed in figure 5.

The vortex structure present immediately prior to the power stroke (figure 5a) included two classes of vortex structures: one vortex was generated from the shear layer at the inner wall (denoted by 1* in figure 5a) and the other was shed from the bell outer wall by the previous recovery stroke (2* in figure 5a). The vortex structure 1* was pinched off into a starting vortex induced by the present power stroke, as depicted by 3* in figure 5(b). The vortex structure 3* was then separated from the bell margin at the end of the power stroke, as shown in figure 5(c). As the recovery stroke began (figure 5d), the stopping vortex was generated by the present recovery stroke. The stopping vortex increased the momentum by interacting with that generated by the outer wall through the forward-moving behaviour (4* in figure 5d) because the vortices had the same rotational direction. The vortex structure 2* from

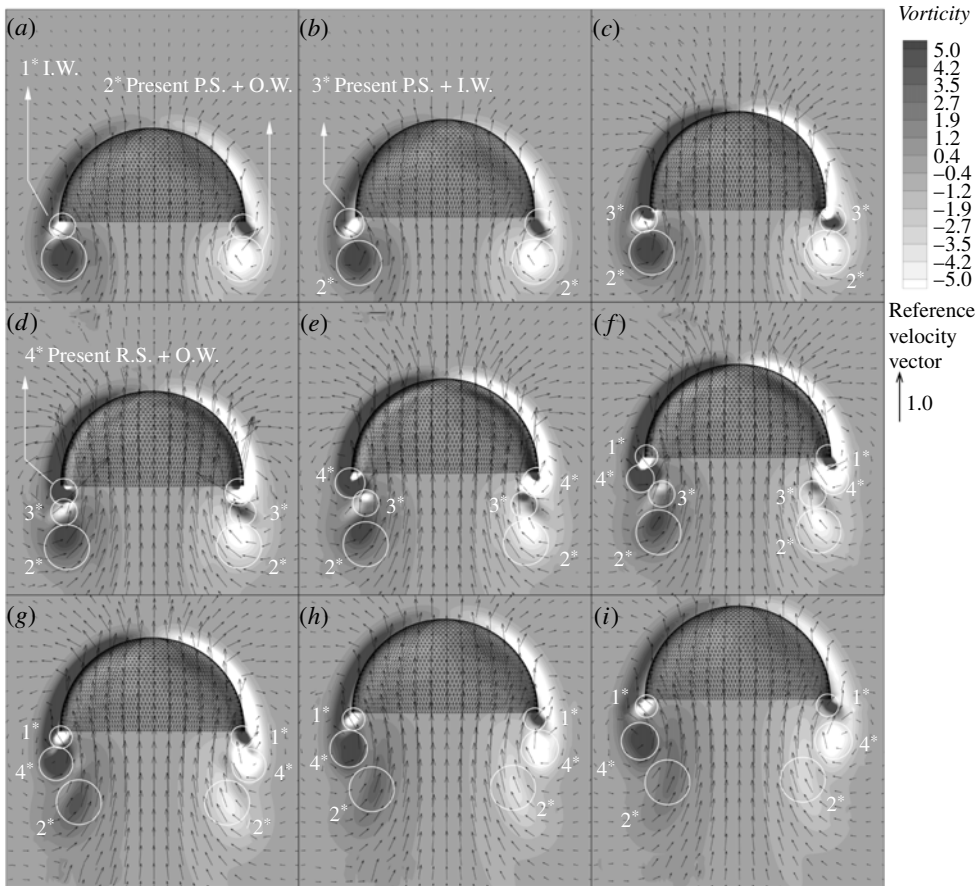


FIGURE 5. The vortex structures and velocity vectors at each instant marked in figure 4(c,d). The structure denoted by 1* is generated from the shear layer at the inner wall (I.W.); 2* is generated by the previous recovery stroke (previous R.S.) and shed from the bell outer wall (O.W.); 3* is generated by the present power stroke (present P.S.) and shed from the bell I.W.; 4* is generated by the present R.S. and shed from the bell O.W. Supplementary movie 1 is available at <http://dx.doi.org/10.1017/jfm.2014.206>. (a) $t = 2.20$; (b) $t = 2.31$; (c) $t = 2.40$; (d) $t = 2.52$; (e) $t = 2.61$; (f) $t = 2.73$; (g) $t = 2.91$; (h) $t = 3.12$; (i) $t = 3.30$.

the previous recovery stroke was generated by the same principle as 4*. As shown in figure 5(d,e), the starting vortex structure 3* dropped from the bell margin played a role in separating the stopping vortex structures 4* and 2*. Vortex cancellation occurred because the starting vortex 3* was surrounded by vortex structures 4* and 2* with the opposite rotational direction; however, the total momentum of the vortex structures 2*, 3* and 4* increased because the vorticity cancellation only occurred at the interface (Dabiri *et al.* 2005). As a result of the interactions between the starting vortex (generated by power strokes) and the stopping vortex (generated by recovery strokes), a large amount of the surrounding fluid was enrolled into the centre axis of the swimming jellyfish, as shown in the velocity vector fields in figure 5. As the recovery stroke proceeded, the vortex structure 4* continued to grow and then was separated from the bell margin, as shown in figure 5(e-g). After

the recovery stroke finished, the starting vortex structure 3* was fully dissipated by the surroundings, as shown in figure 5(g). In figure 5(h,i), the stopping vortex structure 2* from the previous recovery stroke was gradually dissipated; the vortex structure 4* played the same role as 2* in figure 5(a) at the beginning of the next pulse. In addition to enabling the propulsion, the induced motion of the surrounding fluid by the vortex structures also has implications for the feeding mechanism, since the fluid can be advected towards the feeding organisms (tentacles and oral arms) which appear in different locations for different species. Swimming and prey capture occur simultaneously in some types of medusa having oblate bell morphologies (Costello 1992; Costello & Colin 1994, 1995; Ford *et al.* 1997), which means that the swimming mechanism is closely related to the feeding mechanism in these taxa. However, the feeding mechanism is more complicated than the induced fluid motion since other conditions such as the position of feeding organisms or the behaviour of prey should also be taken into account.

The propulsion mechanism that contributed the most to the total thrust was identified by analytically calculating the thrust based on the measured normalized volume of the subumbrella region and the exit velocity. The time history of the normalized volume is shown in figure 6(a), and displays a trend similar to the time-dependent normalized diameter (figure 4c). Figure 6(b) shows the exit velocity v_{ex} , defined as the mean velocity of the fluid at the bell exit relative to the forward centre velocity of the bell. Although the fluid behind the jellyfish appears to mainly flow toward the body, as seen in the velocity vector fields in figure 5, the ejection/injection of fluid from/into the bell is evident by subtracting the forward centre velocity from the velocity field (figure 6b). The absolute values of the exit velocities during the contraction and relaxation phases are similar. The time-dependent exit velocity underwent oscillations after each power and recovery stroke due to the elastic energy stored in the bell during the contraction phase. The thrust from the jet propulsion was obtained analytically, as expressed by

$$T_{jet} = \frac{dm}{dt} |v_{ex}|, \quad (3.1)$$

where T_{jet} stands for the thrust in terms of the jet propulsion and dm/dt indicates the mass flow rate of the exhaust. It should be noted that the direction of the jet propulsion thrust was opposite to that of the time rate of momentum change of the exhaust water expelled from the subumbrella region. The time history of the T_{jet} is shown in figure 6(c), indicating that the thrust was generated during the contraction phase and the drag was formed during the relaxation phase. These observations agree with the report of Megill (2002); however, the velocity increased during the middle relaxation phase, as shown in figure 4, suggesting that the total locomotive force was not derived only from the thrust originating from the exhaust water flow. Comparison of figure 6(c) with figure 6(d) suggests that the thrust due to the jet propulsion had a negligible impact on the total locomotion thrust.

Figure 6(d) shows the total thrust force applied to the whole bell along the moving direction, which was calculated by summing the Lagrangian momentum term F_l added to the solid motion equation (Lai & Peskin 2000). Moreover, it should be noted that the flow behind the jellyfish is mainly towards the body before subtracting the centre velocity of the forward motion, which is more similar to a drag wake. Hence, we also calculated the rate of the momentum that the jellyfish transfers to the surrounding fluid, which is found to be almost equivalent to the total thrust force, as seen in figure 6(d), implying that the flow near the bell margin contributes

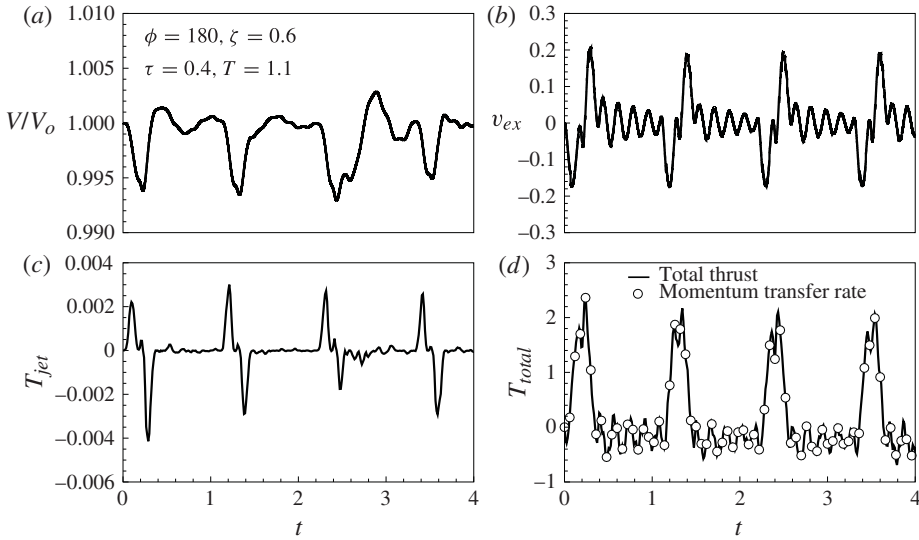


FIGURE 6. Time histories of (a) the normalized volume of the subumbrella region, (b) the velocity relative to the centre velocity at the bell exit, (c) the thrust by the subumbrella volume change, i.e. the jet propulsion, and (d) the total thrust derived from the interaction force with the surrounding fluid and the rate of the momentum that the jellyfish transfers to the surrounding fluid, where the Skalak membrane is characterized by $\phi = 180$ and $\zeta = 0.5$.

significantly to the generation of the thrust force. The rate of the momentum transfer was measured through the whole domain of the fluid since the momentum variation is only caused by the swimming jellyfish. As shown in figure 5, a jellyfish with an oblate bell morphology provides the surrounding fluid with the momentum generating the vortex structures, which contributes to the total thrust to a greater extent. This is supported by Newton's second and third laws, indicating that the locomotive force exerted on the animal by the fluid has an equal magnitude to the rate at which the animal provides the fluid with momentum (Dabiri 2005).

3.2. Effects of the bell elasticity

The power stroke is a result of active muscle contraction and the recovery stroke is a passive behaviour due to the bell elasticity. Jellyfish with different elastic properties were simulated to show the effects of elasticity on the swimming performance of the jellyfish. A tension modulus within the range of 40–280 was selected based on the elastic properties calculated from experimental data and reported elsewhere (Megill 2002). Other parameters, such as the force duration and the period, were set to the values listed above. The time histories of the normalized diameter and volume are plotted in figure 7(a,b) for a series of tension moduli (holding the bending modulus constant) or bending moduli (holding the tension modulus constant), respectively. The time histories of the bell diameter and volume followed similar trends. A jellyfish bell with a low tension modulus deformed to a greater extent than one with a high tension modulus because the rigidity of the jellyfish bell increased as the tension modulus increased. Figure 7(c,d) illustrates the variations in the bell deformation and volume for different bending moduli (holding the tension modulus constant). The deformations

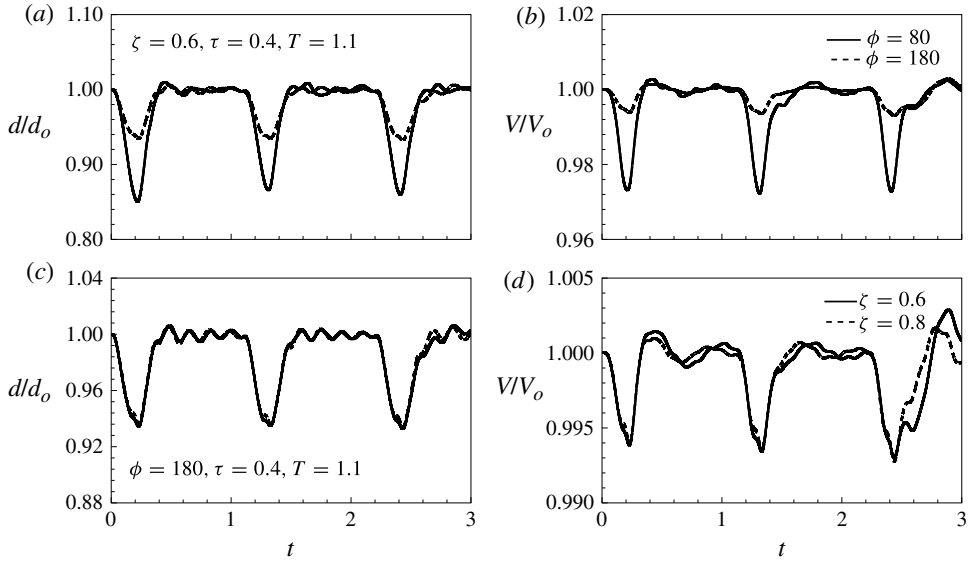


FIGURE 7. Time histories of (a,c) the normalized bell diameter and (b,d) the normalized volume of the subumbrella region, with $\zeta = 0.5$ and various elasticity moduli in (a,b) and $\phi = 180$ and various bending moduli in (c,d).

did not differ significantly, as in the simulations conducted for different tension moduli. Hence, the bell deformation was more affected by the tension modulus than by the bending modulus.

Figure 8(a) shows the average centre velocity v_{avg} of a bell (measured upon reaching the steady state) as a function of the tension modulus and bending modulus. As the bending modulus of the bell increased (holding the tension modulus constant), the average centre velocity decreased slightly. The average centre velocity increased with the tension modulus for low tension modulus values, and then decreased as the tension modulus was further increased. As the elastic modulus decreased, large deformations in the jellyfish bell produced wide vortex structures in the wake that conveyed significant momentum to the ambient fluid; however, the average centre velocity increased as the bell deformations decreased for tension moduli between 40 and 100. Low tension moduli produced such large deformations that the bell oscillated along the direction of motion after the power and recovery strokes. The oscillations generated a reverse flow and exhausted the elastic energy stored during the contraction phase. The average centre velocity decreased as the tension modulus increased beyond 100 because the effects of the energy dissipation via the bell oscillations were less significant. The bell deformations decreased as the tension modulus increased in this range.

Figure 8(b) shows the variations in the Strouhal number as a function of the tension modulus. The Strouhal number is generally defined as $St = f_{fl}A/v_{avg}$, where f_{fl} is the flapping frequency and equals $1/T$, and A is the flapping amplitude. The St number indicates the magnitude of the effects of the flapping frequencies and flapping amplitudes on the centre velocity. In this subsection, the flapping frequency was fixed so that St depended only on the relationship between the flapping amplitude and the centre velocity (because the flapping amplitude was based on the elastic properties of the bell). Hence, St could be interpreted in terms of the ratio of the flapping amplitude

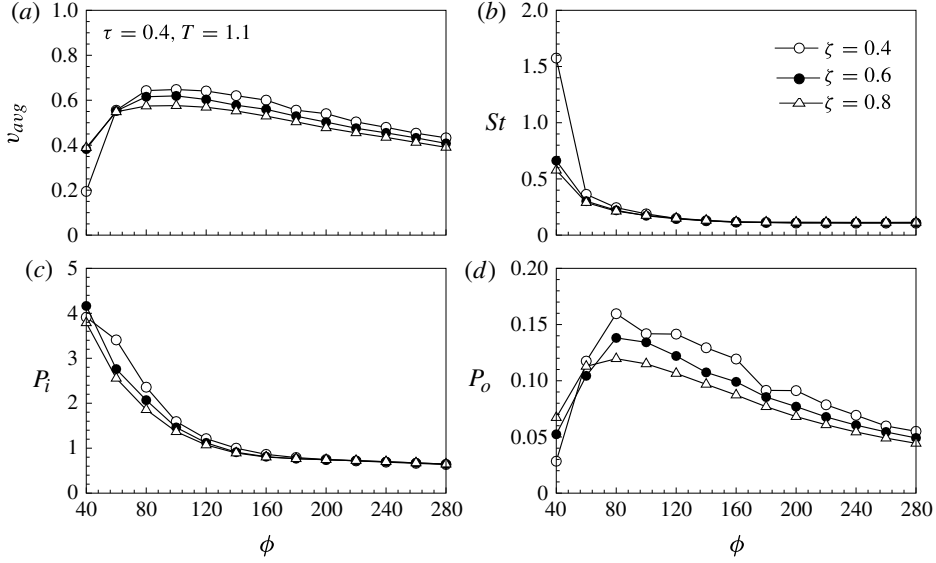


FIGURE 8. Variations of (a) the average velocity during the propulsive cycle at which the centre velocity reaches a steady state, (b) the Strouhal number St , (c) the power input and (d) the power output, with the tension and bending coefficients indicated.

to the distance travelled under the forward motion during one pulse. As shown in figure 8(b), St did not depend significantly on the bending modulus, while it decreased with increment of the tension modulus. Figure 8(b) shows that St approached an asymptotic value beyond a tension modulus of approximately 160, indicating that the distance travelled under the forward motion decreased at almost the same rate as the flapping amplitude decreased. Comparison of the instantaneous vorticity contours with different tension moduli in this range (not shown here) exhibited similar vortex dynamics during the swimming process to that displayed in figure 5, apart from the difference in the strength of the vortex structures.

Figure 8(c,d) shows the power input and output in terms of a mechanical definition. The power input P_i was defined as the energy required to deform the bell divided by the period, according to

$$P_i = f_{fl} \int F_{ex} \frac{dr}{dt} dt, \quad (3.2)$$

where dr/dt is the rate of change of the diameter. The power output P_o was expressed as

$$P_o = v_{avg} F_{th}, \quad (3.3)$$

where F_{th} is the average of the total thrust T_{total} and indicates the interaction force between the swimming jellyfish and the fluid. Figure 8(c) shows that the power input decreased as the tension modulus increased because the deformations were high for low elastic parameter values. As a result, significant energy was required to achieve deformation. The power input varied at low tension moduli but remained relatively constant at high tension modulus values. The trend in the power output, as shown in figure 8(d), resembled the trend observed in the average centre velocity and reached a maximum value at a tension modulus of approximately 80. Similarly, the power output increased as the bending modulus decreased.

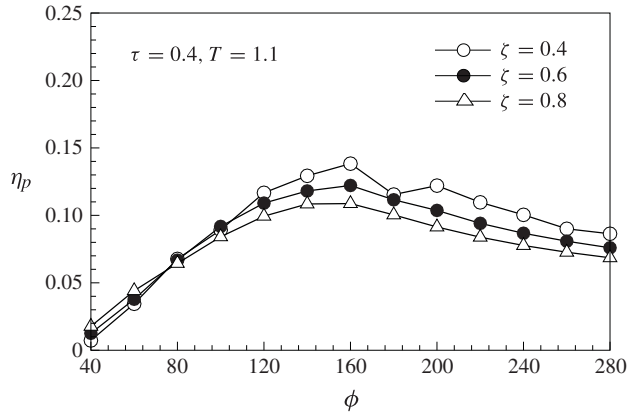


FIGURE 9. Variations in the propulsive efficiency as a function of the tension and bending moduli.

Figure 9 shows the propulsive efficiency η_p as a function of the tension and bending moduli. The propulsive efficiency is defined as the ratio of the power output to the power input and physically indicates the conversion rate of the energy required for bell deformation into the kinematic energy used for travelling. As the bending modulus values increased, the propulsive efficiency decreased slightly. Low tension modulus values required a large amount of energy to deform the bell, and the power output was low due to the effects of the reverse flow. For this reason, the propulsive efficiency increased as the tension modulus increased at low tension values. As shown in figure 9, the propulsive efficiency reached a maximum at a tension modulus of 160, where the optimal energetics is expected. Beyond the optimal tension modulus, the efficiency decreased as the tension modulus increased.

3.3. Effect of the swimming pattern

The behaviour of a swimming jellyfish is affected by the swimming pattern as well as the elasticity of the jellyfish. In this subsection, we examine the effects of the swimming pattern (i.e. variations in the flapping frequency and the force duration) on the swimming performance of a jellyfish with fixed elastic properties corresponding to the parameters used in the simulations described in §3.1. The flapping frequency was determined by the period of the pulse. Figure 10(a) shows the time history of the external force used to contract the bell. The force duration τ and the period T were set to the values used in §3.1, $\tau = 0.4$ and $T = 1.1$. Force was applied intermittently in figure 10(a), whereas a continuous force was applied in figure 10(b). The force duration τ and the flapping period T were given the same value, $\tau = T = 0.4$. Figure 10(c) shows the time histories of the normalized diameter for different flapping periods. In the continuous force case, the bell began a subsequent pulse as the bell reached its maximum extension. On the other hand, under intermittent force conditions, the normalized diameter oscillated after the power and recovery strokes due to the elastic energy stored in the bell. Figure 10(d,e) shows the time histories of the centre position and centre velocity along the direction of motion, indicating that the jellyfish in the continuous force case was able to swim faster than that in the intermittent force case.

Figure 11 shows the vortex contours and velocity vectors for continuous swimming at the specific time marked between the second and fourth strokes in figure 10(c–e).

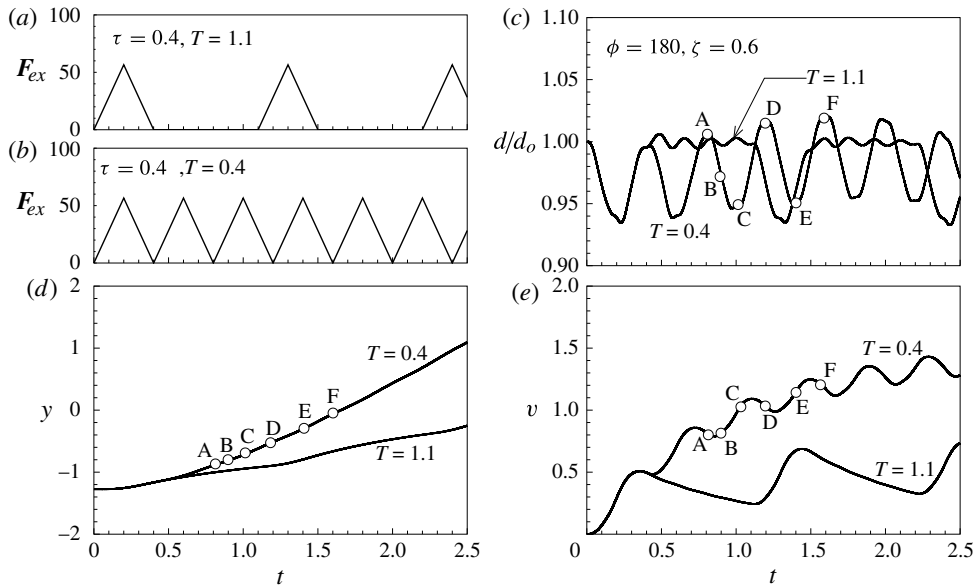


FIGURE 10. Time histories of the body force applied to the bell under (a) intermittent force conditions and (b) continuous force conditions. In (c) the normalized bell diameter, (d) the centre position and (e) the centre velocity are compared for both body force conditions as a function of time. The points marked A, B, C, D, E and F indicate the instants characteristic of the bell deformations during the third to fifth strokes.

The structure 1* (figure 11a) generated from the shear layer at the inner wall was pinched off into the starting vortex by the power stroke (3* in figure 11b). As the contraction proceeded, the vortex structure 3* was shed from the bell margin and dissipated by the surrounding vortex structures (figure 11c). As the recovery stroke started, it was hard to define the concrete structure denoted by 3*, and the stopping vortex formed by the present recovery stroke was merged into the structure that shed from the outer wall, resulting in the vortex structure 4* in figure 11(d). For continuous swimming, unlike the intermittent case, the merged stopping vortex structure in the present pulse (4*) further merged with that formed in the previous pulse (2*), creating an amalgamated vortex structure (2* + 4*), as seen in figure 11(e). After that, the power stroke started without the coasting phase, resulting in vortex structures as shown in figure 11(f). The process of vortex formation depends on the viscosity, because vortex structures are dissipated by the viscosity of the surrounding fluid. Under intermittent force conditions, the vorticity of the vortex structures was mitigated during the coasting period because no momentum was added to the vortex structure. Under continuous force conditions, the pulses were not separated by a time gap, and the continuous strokes produced a vortex structure in the wake with an energy that increased the total momentum of the vortex structure prior to being dissipated by the viscosity. That is to say, the total momentum of the vortex structure increased as the vorticity was augmented by the continuous strokes, resulting in an increased thrust force. A comparison of the vortex contours between the intermittent and continuous swimming cases can be seen in the corresponding animations available online at <http://dx.doi.org/10.1017/jfm.2014.206>.

Several parameters, including the average centre velocity, the distance travelled under the forward motion per stroke cycle, the power input and the power output as

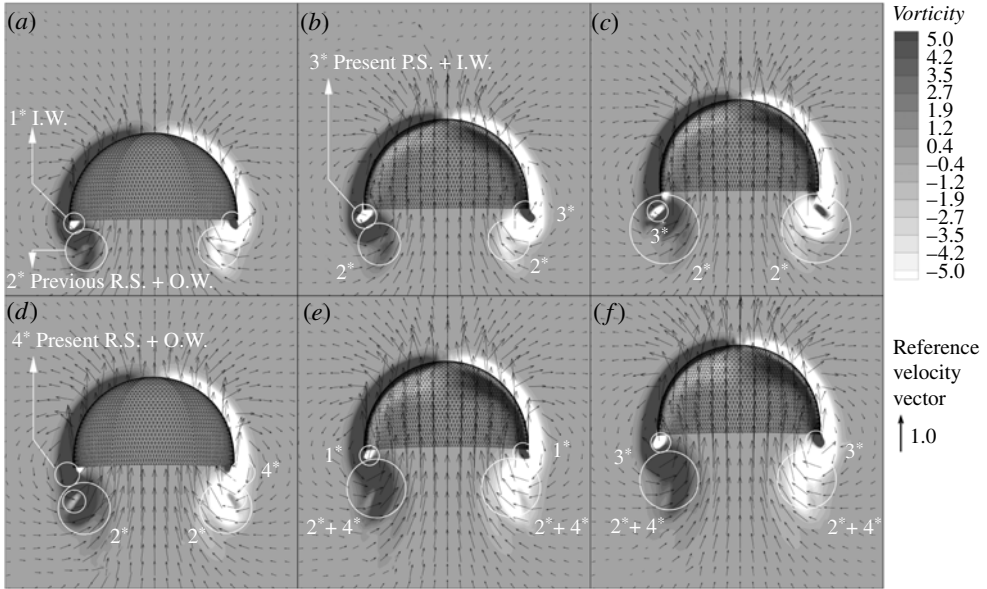


FIGURE 11. The vortex structures and velocity vectors at the instants indicated in figure 10(c–e). The vortex structure 1^* is generated from the shear layer at the I.W.; 2^* is generated by the previous R.S. and shed from the bell O.W.; 3^* is generated by the present P.S. and shed from the bell I.W.; 4^* is generated by the present R.S. and shed from the bell O.W. See supplementary movie 1. (a) $t = 0.81$; (b) $t = 0.90$; (c) $t = 1.02$; (d) $t = 1.11$; (e) $t = 1.20$; (f) $t = 1.29$.

a function of the flapping frequency (f_{fl}), were modelled for fixed elastic properties and force duration (τ), as shown in figure 12. The elastic properties and force duration were set to the values used in the validation studies described above. The average centre velocity during one stroke cycle increased as the flapping frequency increased, as shown in figure 12(a). The distance travelled under the forward motion per stroke cycle did not vary significantly as the flapping frequency increased (figure 12b). Figure 12(c) shows the power input, as defined in (3.2), as a function of the flapping frequency. Under the mechanical definition, the energy required to deform the bell remained constant for a fixed force duration because the body force amplitude and the corresponding flapping amplitude remained unchanged. A high rate of energy transfer was required at high flapping frequencies, yielding the high power input values shown in figure 12(c). Figure 12(d) shows that the power output, defined in (3.3), increased as the flapping frequency increased.

It is estimated that rowing medusae, pulsing their bell margin as a paddle, swim with a high flapping frequency under emergent circumstances such as escape from predators since a high power output is required to accelerate quickly. The power input and power output yielded the propulsive efficiency shown in figure 13 as a function of the flapping frequency. The propulsive efficiency increased as the flapping frequency increased, implying that rowing medusae swim with a high flapping frequency to cover a long distance in which high-efficiency swimming is favourable. However, the rate of energy used for the deformation of the bell (the power input) is large for swimming at a high flapping frequency, as depicted in figure 12(c). For rowing medusae, pulsing their bell margin as a paddle, the rate of energy consumed for

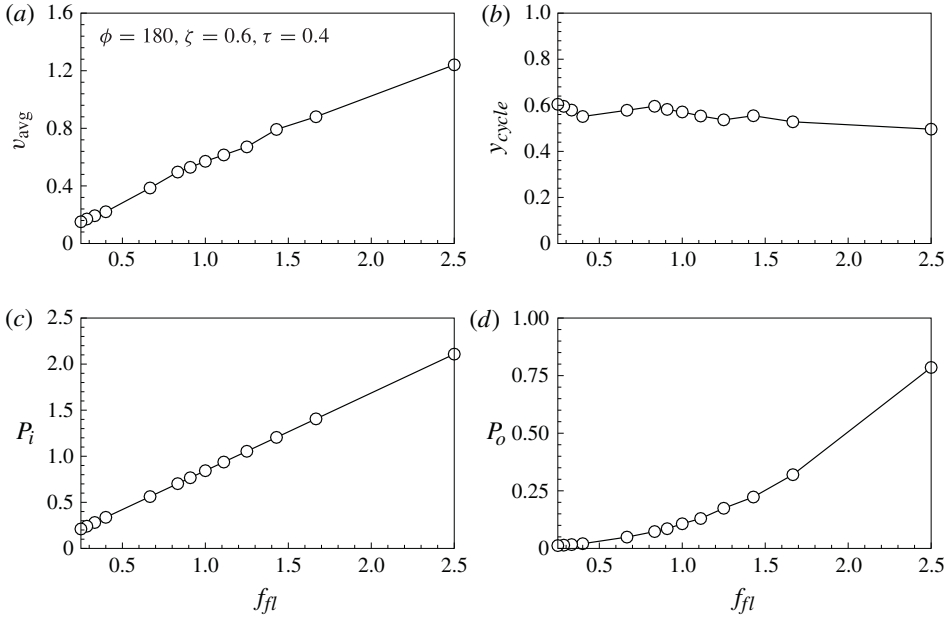


FIGURE 12. Variations in (a) the average velocity during the propulsive cycle at which the centre velocity reached a steady state, (b) the distance travelled during forward motion per stroke cycle, (c) the power input and (d) the power output as a function of the frequency.

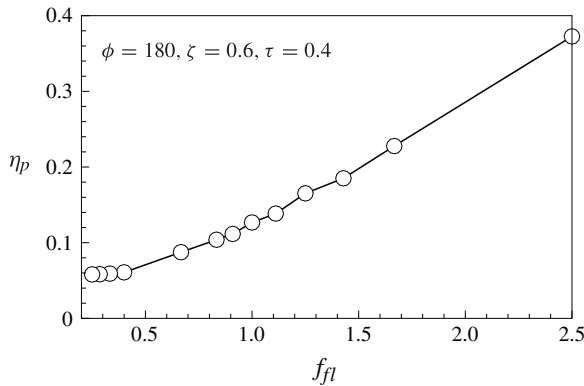


FIGURE 13. Variations in the propulsive efficiency as a function of the flapping frequency.

the bell deformation (power input) is also a significant factor in addition to the propulsive efficiency when they are travelling, since they spend significantly more time in swimming than in resting (Colin *et al.* 2003). For this reason, jellyfish occasionally swim with an intermittent pulse (Megill 2002) when they are migrating. For a given force duration, swimming with a high flapping frequency to achieve a high power output (figure 12d) is favourable to escape from predators; this fortunately has a higher efficiency, as seen in figure 13.

In addition to the flapping frequency, the swimming pattern depends on the force duration. Figure 14(a,b) shows the time histories of the body force required to deform the bell. The force duration and the flapping period were set to $\tau = 0.4$

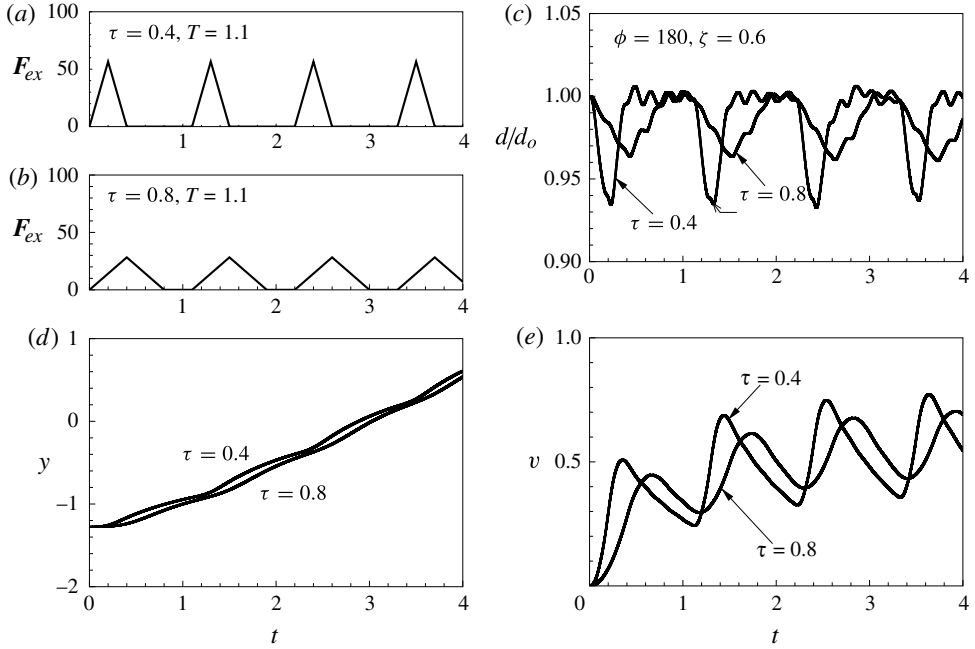


FIGURE 14. Time histories of the body force applied to the bell under intermittent force conditions of (a) $\tau = 0.4$, $T = 1.1$ and (b) $\tau = 0.8$, $T = 1.1$. The (c) normalized bell diameter, (d) centre position and (e) centre velocity variations under each body force condition are compared over time.

and $T = 1.1$, respectively, in figure 14(a). These values were used in the validation study. The values were set to $\tau = 0.8$ and $T = 1.1$ to produce the simulation results shown in figure 14(b). It should be noted that the impulse of the force, defined by the time integral of the force F_{ex} , was the same in both cases. Figure 14(c) shows the time-dependent bell diameter in both cases. The amplitude of the bell diameter variations was large for shorter force durations because the maximum force amplitude was high; however, the bell deformed for a longer period of time under longer force durations. A jellyfish swimming with a shorter force duration but a larger deformation moved slightly faster, as shown in figure 14(d). The time histories of the centre velocity (figure 14e) reveal that the maximum centre velocity during one cycle was higher for a shorter force duration due to the higher amplitude of the body force; however, the centre velocity of a jellyfish swimming with a longer force duration increased over a longer period of time within each stroke cycle, as shown in figure 14(e). The vortex structures generated for the intermittent swimming cases with different forcing durations but the same body force impulse were not significantly different, as shown in supplementary movie 1. The main differences were the length of the period during which momentum was transferred to the vortex structures by the strokes and the vorticity difference caused by the different stroke amplitude. Momentum transfer from the bell to the fluid proceeded over a longer period of time, whereas the vorticity of the structure was lower due to the lower stroke amplitude in the case of the longer force durations.

The average centre velocities remained almost the same, regardless of the force duration, for a fixed flapping frequency, as seen in figure 15(a). As the force duration

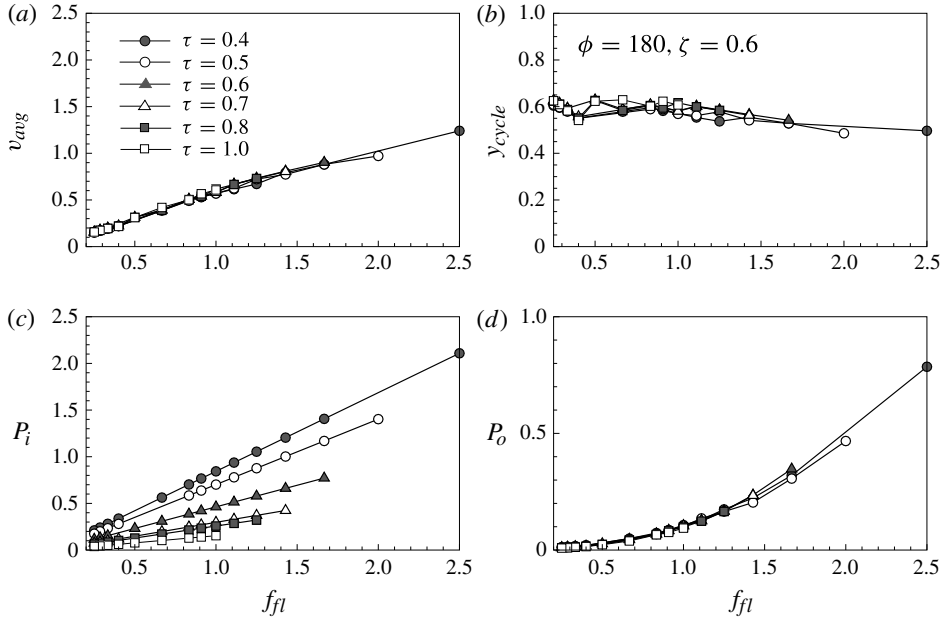


FIGURE 15. (a) The average velocity during the propulsive cycle at which the centre velocity reached a steady state, (b) the distance travelled during forward motion per stroke cycle, (c) the power input and (d) the power output as a function of the flapping frequency and force duration.

increased, the vortex structure generated in the wake received momentum over a longer period of time, thereby compensating for the weakness of the vorticity from the reduced strength of the stroke. Hence, it was predicted that a jellyfish receives the same amount of momentum from the vortex structures for the same impulse of body force exerted on the bell. Figure 15(b) shows the distance travelled during forward motion per stroke cycle as a function of the flapping frequency. The trend is similar to that observed in figure 12(b). At the same flapping frequency, the distance travelled during forward motion is comparable, as shown in figure 14(d). Figure 15(c,d) shows the power input and output for different force durations and flapping frequencies. As shown in figure 14(c), the bell deformed more rapidly and greatly as the force duration decreased, with a correspondingly increased force amplitude (due to the same force impulse), leading to higher power input required to deform the bell. Hence, the power input increased as the force duration decreased. On the other hand, the power output remained nearly constant regardless of the force duration at a given flapping frequency, as shown in figure 15(d).

The flapping amplitude during the first flapping period as a function of the flapping frequency and force duration is plotted in figure 16(a). It is shown that the flapping amplitude remained the same for different flapping frequencies for the same force duration, but decreased for the same flapping frequency as the force duration increased. One should recall that in figure 15(a), the centre velocity has approximately the same value for the same flapping frequency despite the different flapping amplitudes, indicating that other conditions also contribute to the determination of the centre velocity apart from the flapping amplitude and frequency. The reason is the intermittency of the flapping motion of the swimming jellyfish. To

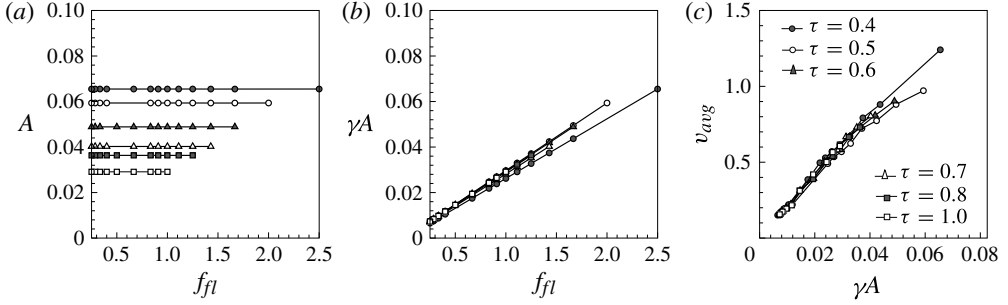


FIGURE 16. Variations of (a) the flapping amplitude and (b) the modulated flapping amplitude with the flapping frequency for different force durations; (c) variations of the average centre velocity with the modulated flapping amplitude for different force durations.

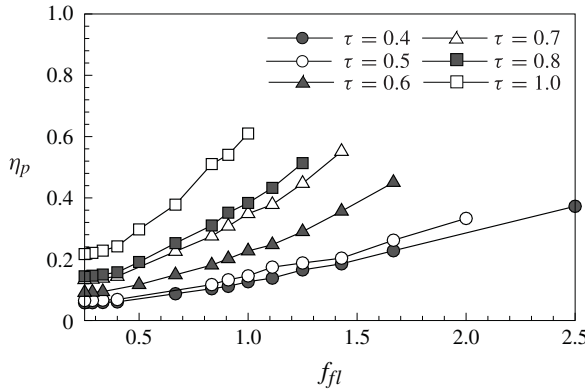


FIGURE 17. Variations in the propulsive efficiency as a function of the flapping frequency and the force duration.

reflect the intermittency in the swimming motion, an intermittency factor γ , defined as the ratio of the force duration τ to the pulse period T , was introduced. For the continuous swimming case, the intermittency factor is 1. Then, we calculated the modulated flapping amplitude defined as the product of the flapping amplitude and the intermittency factor. It is shown in figure 16(b) that the modulated flapping amplitude increased with the flapping frequency, but remained almost the same for different force durations under the condition of the same impulse of the external body force. As seen in figure 16(c), the averaged centre velocity had a good correlation with the modulated flapping amplitude.

The propulsive efficiencies at different flapping frequencies and force durations are shown in figure 17. The propulsive efficiency increased as the force duration increased for a given flapping frequency, as easily derived from figure 15(c,d), given the same impulse of the body force. The propulsive efficiency also increased with increase of the flapping frequency, due to the effect of the added mass. When a body is accelerating or decelerating in a fluid, the body must move some volume of the surrounding fluid, by which the inertia is added to the system. For swimming jellyfish, the intermittent acceleration and deceleration of the fluid surrounding the jellyfish leads to added cost of locomotion (Daniel 1983). In the case of a swimming fish, the effect of added mass is less important since it is able to produce thrust almost

continuously (Megill 2002), which implies that the added mass that the animal carries in a forward motion becomes less significant for a uniform locomotion. For the case of continuous swimming, the acceleration reaction problem is not as significant as for intermittent swimming, in which the coasting phase exists. As a result, despite an increase in the power input, as depicted in figure 15(c), the propulsive efficiency increases as the flapping frequency increases.

4. Conclusions

We have simulated oblate jellyfish using a paddling-based locomotion for swimming through a fluid using a three-dimensional computational model. The shapes of the oblate jellyfish were modelled using a bell shape with a fineness ratio of 0.5. The elastic bell was assumed to be composed of a Skalak membrane with a bending rigidity that mimicked the elastic characteristics of the jellyfish. The interactions between the swimming jellyfish and the surrounding fluid were obtained using the pIBM, in which the solid motion equation and the fluid motion equation were solved separately by adding the momentum force term to each equation. Our computational model was used to investigate the mechanisms underlying oblate jellyfish swimming behaviour. Momentum was transferred to the surrounding fluid by the power and recovery strokes, and the vortex structures generated benefitted locomotion. An important issue was how the elastic bell was forced to move forward with the bell deformation, including the flapping amplitude and frequency, which depended on the elastic properties and the external forces. The effects of the bell elasticity on the swimming performance were investigated. It was found that the bell deformations were significantly affected by the tension modulus but not the bending modulus. The swimming performance was shown to be dependent on the bell elasticity, and an optimal elasticity value was identified that maximized the propulsive efficiency. To examine the effects of the external force on the locomotion in terms of the propulsive efficiency, we simulated and compared a series of cases with different force amplitudes and frequencies. It was found that an oblate jellyfish obtaining thrust by a paddling motion yielded a propulsive efficiency that increased with the force duration given the same impulse of the external body force. The propulsive efficiency also increased with the flapping frequency due to the effect of the added mass. This relationship may differ in other jellyfish species that generate thrust by different mechanisms.

To investigate the mechanism by which the bell deformations drive the forward motion, we visualized the vortex structures, generated in the wake of oblate swimming jellyfish, using a paddling-based locomotion, under different conditions of the flapping period and force duration. The vortex structure from the shear layer at the inner wall was shed into the starting vortex by the power stroke. The stopping vortex was generated by the recovery stroke and merged with the vortex shed from the outer wall. The fluid motion induced by the resulting vortex structures near the bell margin played a significant role in both the locomotion and the feeding mechanism. The vortex structures took a slightly different form depending on the force duration and flapping period, but the distances travelled during one period were indistinguishable, implying that almost the same amount of momentum was transferred to the swimming jellyfish from the vortex structures per cycle for a given impulse of the body force on the bell. We have identified a potential swimming mechanism by which a jellyfish may obtain sufficient locomotive force to cover distances or to escape from predators.

Acknowledgements

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Supplementary movies

Supplementary movies are available at <http://dx.doi.org/10.1017/jfm.2014.206>.

REFERENCES

- ALBEN, S., MILLER, L. A. & PENG, J. 2013 Efficient kinematics for jet-propelled swimming. *J. Fluid Mech.* **733**, 100–133.
- BARTHES-BIESEL, D., DIAZ, A. & DHENIN, E. 2002 Effect of constitutive laws for two-dimensional membranes on flow-induced capsule deformation. *J. Fluid Mech.* **460**, 211–222.
- CIRAK, F., ORTIZ, M. & SCHRODER, P. 2000 Subdivision surfaces: a new paradigm for thin-shell finite-element analysis. *Intl J. Numer. Meth. Engng* **47**, 2039–2072.
- COLIN, S. P. & COSTELLO, J. H. 2002 Morphology, swimming performance and propulsive mode of six co-occurring hydromedusae. *J. Expl Biol.* **205**, 427–437.
- COLIN, S. P., COSTELLO, J. H. & KLOS, E. 2003 *In situ* swimming and feeding behavior of eight co-occurring hydromedusae. *Mar. Ecol. Prog. Ser.* **253**, 305–309.
- COSTELLO, J. H. 1992 Foraging energetics in hydromedusae. *Sci. Mar.* **56**, 185–191.
- COSTELLO, J. H. & COLIN, S. P. 1994 Morphology, fluid motion and predation by the scyphomedusa *Aurelia aurita*. *Mar. Biol.* **121**, 327–334.
- COSTELLO, J. H. & COLIN, S. P. 1995 Flow and feeding by swimming scyphomedusae. *Mar. Biol.* **124**, 399–406.
- COSTELLO, J. H., COLIN, S. P. & DABIRI, J. O. 2008 Medusan morphospace: phylogenetic contrasts, biomechanical solutions, and ecological consequences. *Invertebr. Biol.* **127**, 265–290.
- DABIRI, J. O. 2005 On the estimation of swimming and flying forces from wake measurements. *J. Expl Biol.* **208**, 3519–3532.
- DABIRI, J. O., COLIN, S. P., COSTELLO, J. H. & GHARIB, M. 2005 Flow patterns generated by oblate medusa jellyfish: field measurements and laboratory analyses. *J. Expl Biol.* **208**, 1257–1265.
- DABIRI, J. O., COLIN, S. P., KATIJA, K. & COSTELLO, J. H. 2009 A wake-based correlate of swimming performance and foraging behavior in seven co-occurring jellyfish species. *J. Expl Biol.* **213**, 1217–1225.
- DABIRI, J. O. & GHARIB, M. 2003 Sensitivity analysis of kinematic approximations in dynamic medusa swimming models. *J. Expl Biol.* **206**, 3675–3680.
- DANIEL, T. 1983 Mechanics and energetics of medusan jet propulsion. *Can. J. Zool.* **61**, 1406–1420.
- DANIEL, T. 1985 Cost of locomotion: unsteady medusan swimming. *J. Expl Biol.* **119**, 149–164.
- DEMONT, M. E. & GOSLINE, J. M. 1988a Mechanics of jet propulsion in the hydromedusan jellyfish, *Polyorchis penicillatus*: I, mechanical properties of the locomotor structure. *J. Expl Biol.* **134**, 313–332.
- DEMONT, M. E. & GOSLINE, J. M. 1988b Mechanics of jet propulsion in the hydromedusan jellyfish, *Polyorchis penicillatus*: II, energetics of the jet cycle. *J. Expl Biol.* **134**, 333–345.
- DEMONT, M. E. & GOSLINE, J. M. 1988c Mechanics of jet propulsion in the hydromedusan jellyfish, *Polyorchis penicillatus*: III, a natural resonating bell; the presence and importance of a resonant phenomenon in the locomotor structure. *J. Expl Biol.* **134**, 347–361.
- DULAR, M., BAJCAR, T. & SIROK, B. 2009 Numerical investigation of flow in the vicinity of a swimming jellyfish. *Engng Appl. Comput. Fluid Mech.* **3**, 258–270.
- FORD, M. D. & COSTELLO, J. H. 2000 Kinematic comparison of bell contraction by four species of hydromedusae. *Sci. Mar.* **64**, 47–53.
- FORD, M. D., COSTELLO, J. H., HEIDELBERG, K. B. & PURCELL, J. E. 1997 Swimming and feeding by the scyphomedusa *Chrysaora quinquecirrha*. *Mar. Biol.* **129**, 355–362.

- FUNG, Y. C. & TONG, P. 2001 *Classical and Computational Solid Mechanics*. World Scientific.
- HERSCHLAG, G. & MILLER, L. 2011 Reynolds number limits for jet propulsion: a numerical study of simplified jellyfish. *J. Theor. Biol.* **285**, 84–95.
- HUANG, W.-X., CHANG, C. B. & SUNG, H. J. 2011 An improved penalty immersed boundary method for fluid-flexible body interaction. *J. Comput. Phys.* **230**, 5061–5079.
- HUANG, W.-X., CHANG, C. B. & SUNG, H. J. 2012 Three-dimensional simulation of elastic capsules in shear flow by the penalty immersed boundary method. *J. Comput. Phys.* **231**, 3340–3364.
- HUANG, W.-X. & SUNG, H. J. 2009 An immersed boundary method for fluid-flexible structure interaction. *Comput. Meth. Appl. Mech. Engng* **198**, 2650–2661.
- JOHNSON, T. A. & PATEL, V. C. 1998 Flow past a sphere up to a Reynolds number of 300. *J. Fluid Mech.* **378**, 19–70.
- KIM, K., BAEK, S. J. & SUNG, H. J. 2002 An implicit velocity decoupling procedure for incompressible Navier–Stokes equations. *Intl J. Numer. Meth. Fluids* **38**, 125–138.
- LAI, M.-C. & PESKIN, C. S. 2000 An immersed boundary method with formal second-order accuracy and reduced numerical viscosity. *J. Comput. Phys.* **160**, 705–719.
- LIAO, J. C., BEAL, D. N., LAUDER, G. V. & TRIANTAFYLLOU, M. S. 2003 Fish exploiting vortices decrease muscle activity. *Science* **302**, 1566–1569.
- LIMA, S. L. 1985 Maximizing feeding efficiency and minimizing time exposed to predators: a trade-off in the black-capped chickadee. *Oecologia* **66**, 60–67.
- LINDEN, P. F. & TURNER, J. S. 2004 ‘Optimal’ vortex rings and aquatic propulsion mechanisms. *Proc. R. Soc. Lond. B* **271**, 647–653.
- LIU, H., WASSERSUG, R. J. & KAWACHI, K. 1996 A computational fluid dynamics study of tadpole swimming. *J. Expl Biol.* **199**, 1245–1260.
- MCHENRY, M. J. & JED, J. 2003 The ontogenetic scaling of hydrodynamics and swimming performance in jellyfish (*Aurelia aurita*). *J. Expl Biol.* **206**, 4125–4137.
- MCHENRY, M. J., PELL, A. C. & LONG, J. H. 1995 Mechanical control of swimming speed: stiffness and axial wave form in undulating fish models. *J. Expl Biol.* **198**, 2293–2305.
- MEGILL, W. M. 2002 The biomechanics of jellyfish swimming. PhD thesis, The University of British Columbia.
- MEGILL, W. M., GOSLINE, J. M. & BLAKE, R. W. 2005 The modulus of elasticity of fibrillin-containing elastic fibres in the mesoglea of the hydromedusa *Polyorchis penicillatus*. *J. Expl Biol.* **208**, 3819–3834.
- MILLS, C. E. 1981 Diversity of swimming behaviors in hydromedusae as related to feeding and utilization of space. *Mar. Biol.* **64**, 185–189.
- PATANKAR, S. V. 1980 *Numerical Heat Transfer and Fluid Flow*. McGraw-Hill.
- RUDOLF, D. & MOULD, D. 2010 An interactive fluid model of jellyfish for animation. *Commun. Comput. Inform. Sci.* **68**, 59–72.
- SAHIN, M. & MOHSENI, K. 2009 An arbitrary Lagrangian–Eulerian formulation for the numerical simulation of flow patterns generated by the hydromedusa *Aequorea victoria*. *J. Comput. Phys.* **228**, 4588–4605.
- SAHIN, M., MOHSENI, K. & COLIN, S. P. 2009 The numerical comparison of flow patterns and propulsive performances for the hydromedusae *Sarsia tubulosa* and *Aequorea victoria*. *J. Expl Biol.* **212**, 2656–2667.
- SANTHANAKRISHNAN, A., DOLLINGER, M., HAMLET, C. L., COLIN, S. P. & MILLER, L. A. 2012 Flow structure and transport characteristics of feeding and exchange currents generated by upside-down *Cassiopea* jellyfish. *J. Expl Biol.* **215**, 2369–2381.
- SUMMERS, A. P., DAROUIAN, K. F., RICHMOND, A. M. & BRAINERD, E. L. 1998 Kinematics of aquatic and terrestrial prey capture in *Terrapene carolina*, with implications for the evolution of feeding in *Cryptodire* turtles. *J. Expl Zool.* **281**, 280–287.
- TAYLOR, G. K., NUDDS, R. L. & THOMAS, A. L. R. 2003 Flying and swimming animals cruise at a Strouhal number tuned for high power efficiency. *Lett. Nature* **425**, 707–711.
- ULRIKE, K. M. 2003 Fish ‘n flag. *Science* **302**, 1511–1512.
- ZHAO, H., FREUND, J. B. & MOSER, R. D. 2008 A fixed-mesh method for incompressible fluid–structure systems with finite solid deformations. *J. Comput. Phys.* **227**, 3114–3140.